

Feasibility Analysis

MIMIC-IV QA Benchmark for Coverage vs. Dispersion

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1 Introduction

The experiment’s goal is that *coverage* (facility-location) outperforms *dispersion* (MMR, gMMR, FPS) when the candidate pool of retrieved chunks has a multi-cluster structure. Other QA datasets exist (HotpotQA, 2WikiMultiHopQA, MuSiQue), but these benchmarks have fixed candidate pools that we cannot control.

Ideal Cluster Structure for coverage

1. **Redundant central cluster** - dominant topic, many similar chunks. Top- k over-samples from here. Coverage naturally avoids redundancy.
2. **Complementary cluster(s)** - other relevant aspects of the query. Coverage ensures at least one representative from each.
3. **Niche cluster** - small but informative, not an outlier. Coverage includes it because facility-location rewards covering every region of D .
4. **Outliers** - informative or noisy. Dispersion favors them (maximizes spread); coverage ignores them unless they represent a dense region.

Proposal: build a **custom QA benchmark from MIMIC-IV** where we can (a) verify that candidate pools have the desired cluster structure, (b) control pool composition, and (c) design multi-aspect queries that require many sources.

Part I

MIMIC-IV Dataset

The database is split across two datasets (important CSVs highlighted in blue):

- **MIMIC-IV** $\Rightarrow$ tabular clinical data recording every diagnosis, procedure, and medication for each hospitalization. This can give us **ground truth** for evaluation. There are two submodules:
 - **hosp**: patient tracking (`patients`, `admissions`, `transfers`), diagnoses (`diagnoses_icd`, `d_icd_diagnoses`, `procedures_icd`, `drgcodes`), labs (`labevents`), medications (`prescriptions`, `emar`).
 - **icu**: bedside data (`chartevents`, `inpuvents`, `outpuvents`, `procedureevents`).
- **MIMIC-IV-Note** $\Rightarrow$ free-text clinical notes
 - **discharge**: discharge summaries. Long, narrative clinical documents written by doctors when a patient leaves the hospital.
 - **radiology**: radiology reports.

For our purposes, the critical tables are:

- **discharge** (in MIMIC-IV-Note): $\sim$ 331K discharge summaries.
- **diagnoses_icd**: join table linking ICD-9/ICD-10 (International Classification of Diseases) codes to subject hospitalizations, with approximate importance ordering (`seq_num`). A typical admission has 5-20 ICD codes, often spanning multiple body systems. Definitions for the codes themselves are in the `d_icd_diagnoses` table
- **patients**: gender, `anchor_age`, date of death.
- **admissions**: admission type, insurance, language, marital status, race, discharge location.

All tables link via `subject_id` (patient) and `hadm_id` (hospitalization). The dataset has 331,794 discharge summaries from 145,915 patients. All protected health information (names, dates, locations) has been removed or replaced with placeholders (---).

Some analytics about demographics:

	Hospital admissions	ICU admissions
Number of stays	431,231	73,181
Unique patients	180,733	50,920
Age, mean (SD)	58.8 (19.2)	64.7 (16.9)
Female Administrative Gender, n (%)	224,990 (52.2)	32,363 (44.2)
Hospital length of stay, mean (SD)	4.5 (6.6)	11.0 (13.3)
In-hospital mortality, n (%)	8,974 (2.1)	8,519 (11.6)
One year mortality, n (%)	106,218 (24.6)	28,274 (38.6)
Data period	2008–2019	

2 MIMIC-IV-Note

The notes are of two kinds: **Discharge Summaries** and **Radiology Reports**.

2.1 Discharge Summaries

The structure of the notes seems consistent, with standardized sections:

Section	Content	QA Value
Brief Hospital Course	Clinical reasoning per problem	Highest
History of Present Illness	Symptom narrative, temporal progression	High
Pertinent Results	Labs, imaging, micro interpretations	High
Discharge Diagnosis	Diagnostic labels	High (linkable to ICD)
Chief Complaint	1–2 lines, reason for admission	High only as contextual embedding metadata
Past Medical History	Comorbidities list	Medium
Physical Exam	Admission + discharge findings	Medium
Discharge Medications	Treatment protocols with dosing	Medium
Major Surgical or Invasive Procedure	Procedures performed during stay	Low
Allergies	-	Low
Family History	-	Low
Medications on Admission	-	Low
Discharge Disposition	Destination (home, SNF, rehab, ...)	Low
Discharge Condition	Mental status, consciousness, activity	Low
Discharge Instructions	Patient-facing plain-language summary	Low
Service	Medicine, Surgery, ...	None
Attending	Attending physician	None (deidentified)
Social History	Lifestyle, substance use	None (deidentified)
Followup Instructions	Outpatient follow-up appointments	None (deidentified)

Brief Hospital Course. This section uses # markers to delimit individual clinical problems (e.g., # Stroke, # Hypertension, # DVT). Each sub-section contains *clinical reasoning*: differential diagnoses, treatment rationale, management decisions with justifications. This is where generalizable medical knowledge is embedded within patient-specific narratives.

However, the #-header format is not universal. There are two patterns:

1. **Complex medicine admissions** (the interesting cases): the Hospital Course is organized as a problem list with # headers, one block per active clinical issue. Example: a cirrhosis patient’s Hospital Course has # ASCITES, # HEPATIC ENCEPHALOPATHY, # HYPONATREMIA, # HIV, # COPD (5–8 blocks, each 50–200 tokens). Sometimes there are additional sub-headers like CHRONIC ISSUES or ACUTE/ACTIVE ISSUES grouping the problems. The formatting is inconsistent: # Ascites, # ASCITES, #NSTEMI (with or without space after #, mixed casing).
2. **Simple or surgical admissions** (single-problem cases): the Hospital Course is a single narrative paragraph with *no* # headers. This includes orthopedic (hip fracture), urological (nephrectomy), and straightforward medicine cases (dysphagia workup). These are analogous to single-hop queries. There is only one information peak, so coverage should add nothing.

The TRANSITIONAL ISSUES block. This (sometimes misspelled as TRANISTIONAL ISSUES) appears at the end of many Hospital Courses and contains follow-up action items (“recheck electrolytes”, “follow up with hepatology”), not clinical narratives. These are task-oriented and do not describe clinical problems.

Deidentification artifacts. The ___ placeholder is everywhere in dates, names, ages, and locations. Brief Hospital Course and Physical Exam can contain few of them but largely are usable. Ages are recoverable from `anchor_age` in the `patients` table.

2.2 Radiology Reports

The `radiology` table in MIMIC-IV-Note contains ~2.3M radiologist reports spanning x-ray, CT, MRI, and ultrasound. Reports follow a structured format with sections: **Examination**, **Indication**, **Technique**, **Comparison**, **Findings**, and **Impression**. The associated `radiology_detail` table provides exam codes and exam names (e.g., CT HEAD W/O CONTRAST, LIVER OR GALLBLADDER US) that can be used for filtering.

Content quality is dual:

- **Formulaic reports** (routine chest x-rays, follow-up films): short, low information content. Typical impression: *“No acute cardiopulmonary abnormality.”*
- **Detailed diagnostic reports** (CT head, CT abdomen with contrast, MRI brain, CT angiography): substantial findings sections with organ-by-organ descriptions, measurements, and clinical correlation. Example from a CT head: *“Subcortical and periventricular white matter hypodensities consistent with chronic small vessel ischemic disease. No evidence of acute intracranial hemorrhage, edema, mass effect, or large territorial infarction.”*

Potential value for coverage vs. dispersion. For imaging-dependent conditions (stroke, PE, pneumonia), radiology reports provide a **different semantic perspective** from discharge notes: the discharge note describes clinical management (“tPA administered within 3h window”), while the radiology report describes anatomical findings (“acute hypodensity in L MCA territory, no midline shift”). This seems good for coverage.

Redundancy with discharge notes. The “Pertinent Results” section of discharge summaries often summarizes key imaging findings. Radiology reports may not add unique facets beyond what is already captured there.

I think Discharge Notes are better as a starting point and this can be used later if needed.

2.3 Considerations

2.3.1 Patient-specific vs. Generalizable data

Discharge notes are patient-specific narratives. Meanwhile, the target queries ask for generalizable medical knowledge:

“Quali sono i sintomi dell’ictus in generale e cosa cambia se il paziente ha già avuto un infarto?”

The mismatch is what makes MIMIC interesting for coverage. Top- k by similarity will over-retrieve notes describing the most common presentation (central cluster). **Coverage** will ensure representation from atypical presentations, comorbidity interactions, and edge cases.

3 MIMIC-IV Structured Data

3.1 Hosp Module

The **hosp** module captures hospital-wide data from admission to discharge. Key table groups (tables [highlighted in blue](#) are useful for us):

Patient tracking. Three tables describe demographics and movement:

- [patients](#): gender, [anchor_age](#) (deidentified age), date of death. Ages are shifted but intervals are preserved, so age *at admission* is recoverable.
- [admissions](#): admission/discharge times, admission type (emergency, elective, urgent), insurance, language, marital status, race, discharge location, in-hospital mortality flag.
- [transfers](#): ward-level ADT (admission–discharge–transfer) records tracking physical location throughout the stay.

Billing and diagnoses.

- [diagnoses_icd](#): up to 39 ICD-9-CM / ICD-10-CM codes per admission with approximate importance ordering ([seq_num](#)). Definitions in [d_icd_diagnoses](#). This is our primary source for condition-level labels and query scaffolding.
- [procedures_icd](#): billed procedures (ICD-9-PCS / ICD-10-PCS). Definitions in [d_icd_procedures](#).
- [drgcodes](#): Diagnosis Related Group codes (AP-DRG and MS-DRG) assigning an overall cost category to each hospitalization.
- [hcupsevents](#): HCPCS billing codes for provided services (e.g. mechanical ventilation, ICU care). Definitions in [d_hcupcs](#).

Measurements.

- [labevents](#): laboratory results (chemistry, hematology, blood gases, etc.) with reference ranges and flags. Concept definitions in [d_labitems](#). Multiple measurements may share a [specimen_id](#) (same blood draw).
- [microbiologyevents](#): microbiology cultures with specimen type, organism, antibiotic sensitivities, and dilution results. Stored as a directed hierarchy (specimen → organism → antibiotic).
- [omr](#): Online Medical Record outpatient measurements - blood pressure, height, weight, BMI, eGFR - available from both inpatient and outpatient visits.

Medications. Four tables track prescription and administration:

- [prescriptions](#) + [pharmacy](#): provider orders with drug name, dose, route, and formulary details.
- [emar](#) + [emar_detail](#): electronic Medicine Administration Record (barcode-scanned actual administrations). One row per administration event, linked to [pharmacy](#) and [poe](#) tables.

Administration. `services` (admitting service), `poe` and `poe_detail` (provider order entry - orders for imaging, consults, nutrition, medications).

3.2 ICU Module

The `icu` module (73,181 ICU stays, 50,920 unique patients) contains bedside data from the MetaVision clinical information system (iMDsoft), limited to patients admitted to an ICU. All events reference a `stay_id` (unique per ICU stay), defined in `icustays` with admission/discharge times and care unit names (e.g. MICU, SICU, CCU).

Event tables. Each documented item is an “event” grouped by data type:

Table	Content	Data Type
<code>chartevents</code>	Vital signs, GCS, vent settings, nursing assessments	Numeric time-series
<code>inpuvents</code>	IV infusion rates/amounts (vasopressors, sedation)	Numeric
<code>outpuvents</code>	Urine output, drain volumes	Numeric
<code>procedureevents</code>	Ventilation, CRRT, catheters (start/end times)	Categorical + timestamps
<code>datetimeevents</code>	Date-valued events (e.g., last dialysis)	Dates
<code>ingredientevents</code>	Ingredient-level breakdown of infusions	Numeric
<code>d_items</code>	Dimension table for <code>itemid</code>	Reference

Suitability for embeddings. ICU data is predominantly **numerical time-series** (vital signs, infusion rates, urine output) rather than natural language. These signals are not directly embeddable with sentence transformers, and converting them to text (e.g., “heart rate 92 bpm at 14:30”) would produce low-information chunks unlikely to match condition-level queries semantically. The ICU module is not a primary source for our corpus.

Metadata enrichment. ICU data could strengthen the **contextual embedding prefix** (Section 4) for patients who had ICU stays. The `mimic-code` repository already provides SQL concepts for severity scores (SOFA, SAPS-II, OASIS), ventilation status, and vasopressor use. A richer prefix:

Discharge note for elderly male, admitted for sepsis
(SOFA 14, mechanically ventilated, on vasopressors).
Comorbidities: CHF, CKD.
Section: Brief Hospital Course.

is more informative than one without ICU severity signals. However, this enrichment only matters for **ICU-relevant conditions** (sepsis, respiratory failure, shock, post-cardiac arrest) where severity scores carry discriminative information. For routine admissions (e.g. elective surgery, uncomplicated pneumonia) it may be doesn’t make sense.

Part II

QA Dataset Generation Pipeline

4 Pipeline draft

Phase 1: Corpus Construction

1. **Select target conditions.** Query `diagnoses_icd` for ICD code groups with ≥ 200 (TBD based on dataset statistics. TODO: gather analytics) admissions.

Candidate conditions: stroke, myocardial infarction, pneumonia, sepsis, heart failure, COPD exacerbation, acute kidney injury, pulmonary embolism, diabetic ketoacidosis, cirrhosis, etc. Target: 50–100 conditions.

2. **Build per-condition candidate pools.** For each condition, gather all discharge note chunks from admissions with that ICD code. This is the full candidate set D for facility-location.
3. **Chunk by section + sub-problem.**
 - Split on section headers
 - Within Brief Hospital Course, split on # problem markers
4. **Attach metadata to each chunk:**
 - `subject_id`, `hadm_id`, section name
 - All ICD codes for that admission
 - Demographics: age group, gender, race, insurance
 - Admission type, discharge disposition
5. **Deduplicate same-patient chunks.** The same patient can have multiple hospitalizations with nearly identical boilerplate sections, e.g., **Past Medical History**, **Allergies**, and **Family History** are often copied verbatim across admissions.

Strategy: compute a content hash (e.g., SHA-256 of normalized text) per chunk and deduplicate by (`subject_id`, section, hash). Check also if it's worth considering *only* the most up to date note for a certain patient for a subset of repetitive sections (Past Medical History, etc.)

Phase 2: Embeddings Generation

Raw chunks are too patient-specific to match condition-level queries. Following the **Contextual Retrieval** approach [3], prepend a metadata prefix to each chunk before embedding:

```
[age_group] [gender] [race] patient admitted for [primary ICD description].  
Chief complaint: [chief_complaint].  
Comorbidities: [top ICD descriptions].  
Section: [section_name].
```

Check out Section 6 for more details.

The specific contents to include are to be decided, based on the queries we want to build.

Phase 2.5: Cluster Structure Validation

Before generating queries, verify that per-condition candidate pools exhibit the multi-cluster structure that makes coverage meaningful. Run the UMAP+HDBSCAN pipeline described in Section 7 on each condition's embedded chunks. Check out the section for details and caveats

Phase 3: Query Generation

1. Pick a primary condition (e.g., stroke, ICD-10: I63.*)
2. Pick 1–2 **modifier axes** from metadata:
 - Comorbidity: prior MI, diabetes, renal failure
 - Demographic: elderly, women, specific race
 - Complication: post-surgical, recurrent, with hemorrhagic transformation
3. Prompt an LLM:

“Generate a medical question that requires knowing both the general presentation of [condition] AND how it differs when [modifier]. The answer should require information from multiple patient cases.”

4. Gold answer generation

The idea: give an LLM access to the candidate pool, have it answer the query comprehensively, and for each piece of information in the answer **cite which chunk(s)** it drew from. The cited chunks become the **gold set**, each tagged with a facet label.

Facet annotation. Each gold chunk receives a `facet_label` (e.g., `general_symptoms`, `cardiac_confounding`, `treatment_difference`). These labels enable the **aspect recall** metric (Section 4).

Gold annotation via map-reduce. Since the corpus is heavily pruned before this stage (only conditions that pass the cluster gate, only notes with the relevant ICD codes), each per-condition candidate pool should not be too large. However, to face the long-context that the candidates may give rise to, we can:

- (a) **Map:** split the pool into batches of ~ 50 chunks ($\sim 10K$ tokens). For each batch, the LLM extracts facts relevant to the query and cites which chunk(s) each fact came from.
- (b) **Reduce:** a merge pass synthesizes across batches, deduplicates facets, and produces the final gold answer with citations. Each cited chunk receives a facet label.

5. (Optional) Query quality verifier

- A separate reader-LLM attempts to answer from baseline top- k chunks
- If it succeeds $\Rightarrow$ query does *not* require diversity $\Rightarrow$ discard
- If it fails but succeeds with gold chunks $\Rightarrow$ query requires diversity $\Rightarrow$ keep

Phase 4: Evaluation

- **Aspect recall:** fraction of annotated facets covered by selected k chunks.

$$AR(S) = |\{f \in F : \exists c \in S \text{ with facet } f\}| / |F|$$
- **Fac score:** facility-location objective value $g(S) = \sum_{i \in D} \max_{j \in S} w_{ij}$
- **Jaccard:** overlap between method’s selection and top- k baseline
- **DocRec / FactRec:** gold document/fact fraction recovered

Methods to compare: top- k (baseline), MMR, gMMR, facility-location

5 LLM Usage and Data Use Agreement

Phases 3–4 require feeding MIMIC discharge note chunks into an LLM (query generation, gold answer annotation, optional quality verification). The PhysioNet Credentialed Health Data License 1.5.0 and DUA 1.5.0 impose two constraints:

1. **No sharing.** *“I will not share access to PhysioNet restricted data with anyone else.”* Sending clinical text to a commercial API arguably violates this clause.
2. **Security.** *“I will exercise all reasonable and prudent care to maintain the physical and electronic security of PhysioNet restricted data.”* Even with zero-retention API tiers, the data still transits and is processed on third-party servers outside our control.

Local inference. All LLM steps that touch patient data should run on a self-hosted model. Available hardware: NVIDIA L4 (23 GB VRAM). Speed? Performance of small LLMs?

Exception. Steps that use only *public* information (ICD code descriptions, condition names, modifier labels) can use commercial APIs, since no MIMIC text is transmitted. This applies to query template generation (Phase 3, step 3) but not to gold annotation.

6 Embeddings: considerations

6.1 Contextual Embeddings

A raw chunk from Brief Hospital Course might read:

“Pt had L-sided weakness, slurred speech, presented within 3h window. tPA administered. CT head negative for hemorrhage.”

Without context, embedding this chunk places it in a generic “neurological emergency” region. A query about “stroke symptoms” may not rank it highly, because the semantics is different.

Contextual Retrieval [3] Prepend contextual information to each chunk before embedding. Reported: 49% reduction in retrieval failure rate (combined with BM25 hybrid).

The original method uses an LLM to generate the prefix from the surrounding document, since document structure is unknown in the general case. MIMIC discharge notes, however, come with structured metadata (ICD codes, demographics, admission details) already parsed in relational tables, so we replace the LLM call with a deterministic template.

6.2 Chunking vs. Whole-Document Embedding

Long-context embedding models (e.g., BioClinical ModernBERT with 8K tokens, or Jina v2 with 8K tokens) can embed an entire discharge note as a single vector. Should we consider?

A whole-document embedding of a 2,000-token discharge note compresses heterogeneous content (chief complaint, clinical reasoning,...) into a single 768-dim vector. Mean-pooling over these diverse sections produces a vector that sits in a “generic admission” region of the embedding space, rather than near any specific clinical concept. This flattens the multi-cluster structure we need, so coverage brings no advantage.

When whole-document embedding? If queries target *patient-level* distinctions rather than *information-level* distinctions (e.g., “find patients similar to this case” rather than “what are the symptoms of X and how do they differ with Y”), whole-document embedding is more natural. This is based upon the queries we want to build, so maybe it’s good

6.3 Embedding Models for medical domain

- **BioLORD-2023 [10]:** sentence-transformer fine-tuned from `all-mpnet-base-v2`. 768-dim embeddings, 512-token context.
- **MedEmbed [4]:** family of sentence-transformers (small/base/large) fine-tuned from the BAAI/BGE series. The large variant produces 1024-dim embeddings, 512-token context. Apache-2.0 licence.
- **BioClinical ModernBERT Embeddings [11]:** sentence-transformer fine-tuned from BioClinical ModernBERT, a ModernBERT variant further pre-trained on PubMed abstracts and clinical text in the MIMIC lineage. 768-dim embeddings, up to 8,192-token context. Apache-2.0 licence.

7 Cluster Validation

Do candidate pools have the desired multi-cluster structure of Section 1?

7.1 Pipeline draft

1. **Embed** all chunks with `multi-qa-mpnet-base-cos-v1` (768 dimensions).
2. **UMAP** reduction to $\sim 15\text{--}30$ dimensions. UMAP [8] preserves local *and* global structure better than t-SNE.
3. **HDBSCAN** [5] with `min_cluster_size` tuned per condition.
4. **Validate** structure with:
 - **DBCV** [9]: density-based cluster validation, designed for HDBSCAN output
 - **Cluster size ratio**: $|C_{\max}|/|C_{\text{total}}|$ measures dominance of central cluster
 - **Density ratio**: $\rho_{\text{central}}/\rho_{\text{niche}}$
 - **Number of clusters + noise fraction**
5. **Cross-reference** HDBSCAN clusters with ICD sub-codes. If embedding clusters align with clinical sub-types, the structure is semantically meaningful.

7.2 Selection Criteria

- **Include** conditions where HDBSCAN finds ≥ 3 clusters with DBCV above a threshold. Expected for complex conditions
- **Exclude** conditions with single-cluster pools not relevant for coverage

7.3 High-Dimensional Clustering: Caveats

Curse of dimensionality. HDBSCAN relies on point density. In high dimensions, pairwise distances concentrate [1] and density-based clustering is unreliable. This is why the UMAP + HDBSCAN pipeline (used by BERTopic [7], Top2Vec [2]) is standard practice. Methods like PCA or t-SNE don't perform as well for the dimensionality reduction step.

UMAP is not perfect. UMAP is stochastic and sensitive to hyperparameters (`n_neighbors`, `min_dist`). Results should be validated across multiple UMAP settings.

Anisotropy of embedding spaces. Ethayarajh [6] showed that embeddings occupy an *anisotropic cone*. They do not fill the space uniformly. Clusters may reflect frequency, genre, or style artifacts rather than the semantic distinctions we care about.

Do the four ideal cluster types map to geometric clusters? Idk maybe this is overengineering, maybe there is a simpler way or maybe this step is useless

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